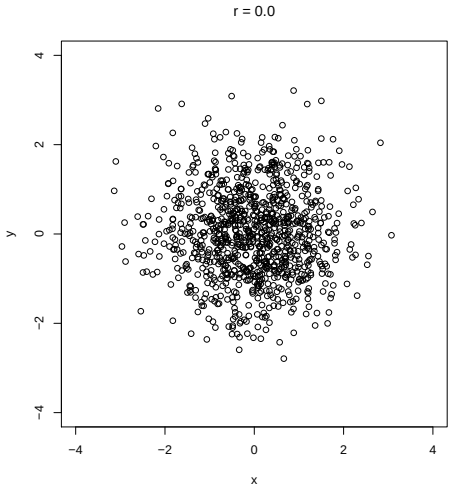
AES519: Covarianza y correlación

Felipe Osorio

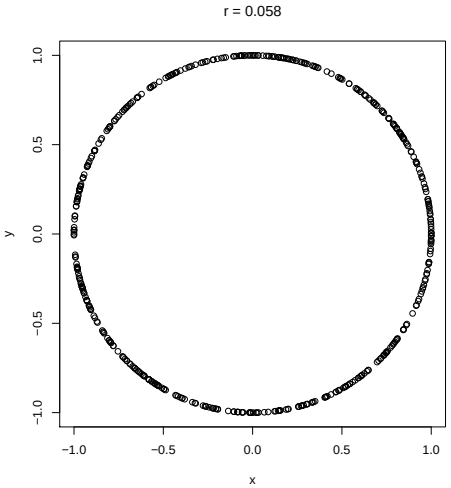
https://faosorios.github.io

Instituto de Matemática Fisica y Estadística, UDLA

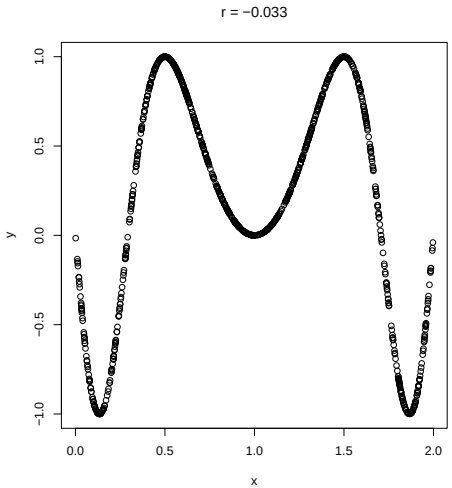
Correlación: Midiendo asociación lineal



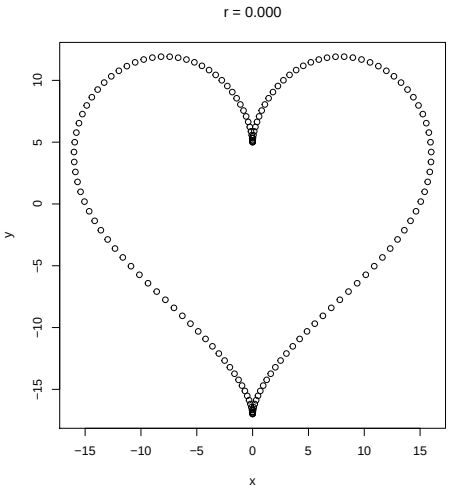
(a) $r = 0{,}000$



(b) $r = 0{,}058$

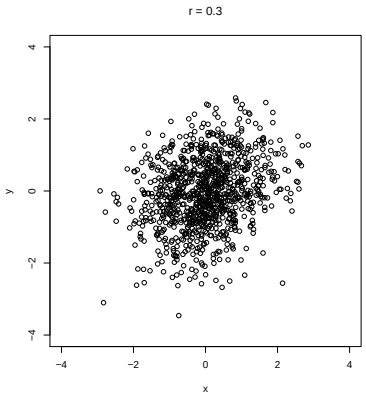


(c) $r = -0{,}033$

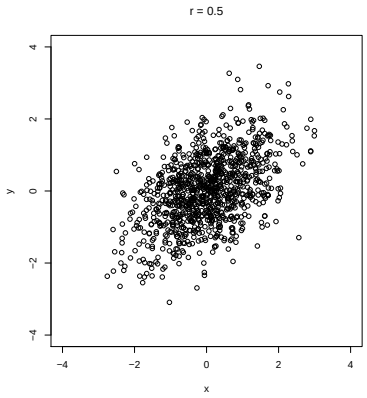


(d) $r = 0{,}000$

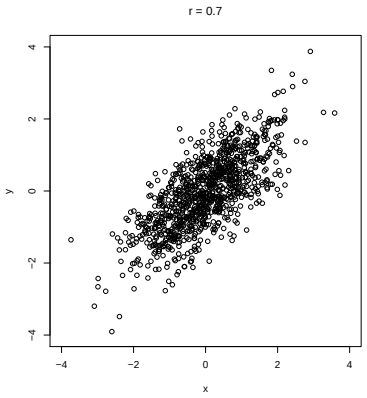
Correlación: Midiendo asociación lineal



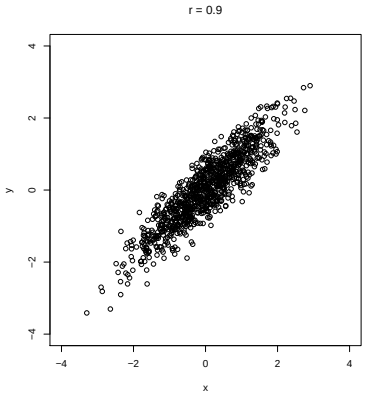
(a) $r = 0{,}30$



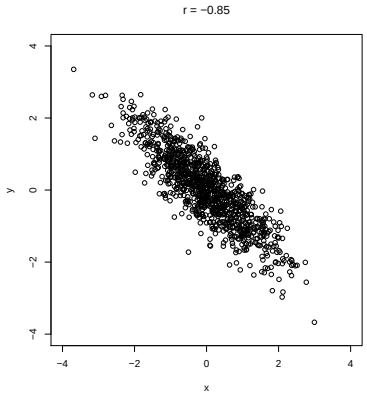
(b) $r = 0{,}50$



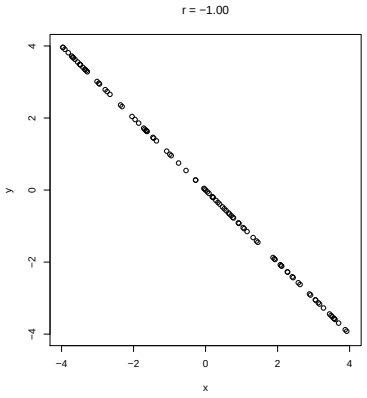
(c) $r = 0{,}70$



(d) $r = 0{,}90$



(e) $r = -0{,}85$



(f) $r = -1{,}00$

Covarianza

Definición 1 (Covarianza):

Para el conjunto $(x_1, y_1), \ldots, (x_n, y_n)$, se define la covarianza como una medida de variabilidad conjunta de dos variables cuantitativas, como:

$$\mathsf{cov}(\boldsymbol{x}, \boldsymbol{y}) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \overline{x})(y_i - \overline{y}).$$

Observación:

Evidentemente, $\mathsf{cov}(\boldsymbol{x}, \boldsymbol{x}) = \mathsf{var}(\boldsymbol{x}) = s_x^2$.

Propiedades de la covarianza

Propiedades:

(a)

$$\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y}) = \sum_{i=1}^{n} x_i y_i - n\overline{x}\,\overline{y}.$$

(b)

$$\mathsf{cov}(a\boldsymbol{x} + b, c\boldsymbol{y} + d) = ac\,\mathsf{cov}(\boldsymbol{x}, \boldsymbol{y}).$$

Definición 2 (Correlación):

La correlación entre $\boldsymbol{x} = (x_1, \ldots, x_n)^\top$ e $\boldsymbol{y} = (y_1, \ldots, y_n)^\top$ es la covarianza de sus versiones estandarizadas. Es decir,

$$\begin{aligned}
\mathsf{cor}(\boldsymbol{x}, \boldsymbol{y}) &= \frac{1}{n-1} \sum_{i=1}^{n} \left(\frac{x_i - \overline{x}}{s_x} \right) \left(\frac{y_i - \overline{y}}{s_y} \right) \\
&= \frac{\sum_{i=1}^{n} (x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum_{i=1}^{n} (x_i - \overline{x})^2 \sum_{i=1}^{n} (y_i - \overline{y})^2}} \\
&= \frac{\mathsf{cov}(\boldsymbol{x}, \boldsymbol{y})}{\sqrt{\mathsf{var}(\boldsymbol{x})\, \mathsf{var}(\boldsymbol{y})}}.
\end{aligned}$$

Observación:

$\mathsf{cor}(\boldsymbol{x}, \boldsymbol{y})$ es una medida adimensional.

Propiedades de la correlación

Propiedades:

(a)
$$\mathsf{cor}(a\boldsymbol{x} + b, c\boldsymbol{y} + d) = \pm\,\mathsf{cor}(\boldsymbol{x}, \boldsymbol{y}).$$

(b)
$$\mathsf{var}(\boldsymbol{x} + \boldsymbol{y}) = \mathsf{var}(\boldsymbol{x}) + \mathsf{var}(\boldsymbol{y}) + 2\,\mathsf{cov}(\boldsymbol{x}, \boldsymbol{y}).$$

(c)
$$\{\mathsf{cor}(\boldsymbol{x}, \boldsymbol{y})\}^2 \leq 1.$$

Observación:

- Evidentemente, $-1 \leq \mathsf{cor}(\boldsymbol{x}, \boldsymbol{y}) \leq 1$.
- Cuando $\mathsf{cor}(\boldsymbol{x}, \boldsymbol{y}) = 0$, diremos que $\boldsymbol{x}$ e $\boldsymbol{y}$ son no correlacionados.

Definición 3 (Coeficiente de correlación de Spearman):

Suponga los datos pareados $(x_1, y_1), \ldots, (x_n, y_n)$. Sea R_i, S_i los rangos de x_i e y_i, respectivamente $(i = 1, \ldots, n)$. Entonces el coeficiente de correlación de Spearman es dado por

$$r_{\mathsf{S}} = \frac{\sum_{i=1}^{n}(R_i - \overline{R})(S_i - \overline{S})}{\sqrt{\sum_{i=1}^{n}(R_i - \overline{R})^2 \sum_{i=1}^{n}(S_i - \overline{S})^2}}.$$

Observación:

Sea

$$D = \sum_{i=1}^{n}(R_i - S_i)^2,$$

y suponga que no existen empates entre los x's e y's, entonces podemos escribir

$$r_{\mathsf{S}} = 1 - \frac{6D}{n^2 - 1}.$$

Ejemplo: Distorsiones de Lenna[1]

(a) $r = 1{,}000$

(b) $r = 0{,}903$

(c) $r = 0{,}991$

(d) $r = 0{,}915$

(e) $r = 0{,}983$

(f) $r = 0{,}799$

[1] (a) Original, (b) sal y pimienta, (c) filtro mediana, (d) ruido speckle, (e) filtro Lee, (f) imagen saturada.

Ejemplo: Distorsiones de Lenna

```
> library(SpatialPack)
# https://github.com/faosorios/SSIM/blob/master/data/lena.rda
> load("lena.rda") # carga datos de Lena

# aplica distorsiones y filtros
> lena.05 <- clipping(lena, low = 0.5) # saturación
> lena.sp <- imnoise(lena, type = "saltnpepper")
> lena.speckle <- imnoise(lena, type = "speckle")
> lena.med <- denoise(lena.sp, type = "median") # filtro mediana
> lena.lee <- denoise(lena.speckle, type = "Lee") # filtro de Lee

# calculando correlaciones
> x <- as.vector(lena) # 262144 observaciones
> cor(x, x)
[1] 1
> cor(x, as.vector(lena.05))
[1] 0.7997093
> cor(x, as.vector(lena.sp))
[1] 0.9028631
> cor(x, as.vector(lena.med))
[1] 0.9907281
> cor(x, as.vector(lena.speckle))
[1] 0.9154696
> cor(x, as.vector(lena.lee))
[1] 0.9829129
```